

Solidity and Smart Contract Development

Homework 1

No AI was used in the writing of the code or the document

GitHub Pull requests.

I commented Gabriel's Pull Request, as I was never added as verifactor.

Question 1 - Press the "Check MetaMask Balance" button with your Metamask connected. How much Wei does it say you have and is this correct? Give a short reason for your belief. (1 point)

Once the metamask button is clicked, the website shows that I have zero Wei on my account, which is correct on the ethereum mainnet. This is explained as the function request from my web browser `eth_getBalance`, with the `currentAccount` parameters. However, I don't have ethereum on the mainnet, only on the Sepolia testnet. This is therefore why I don't have any Wei showing on the webpage

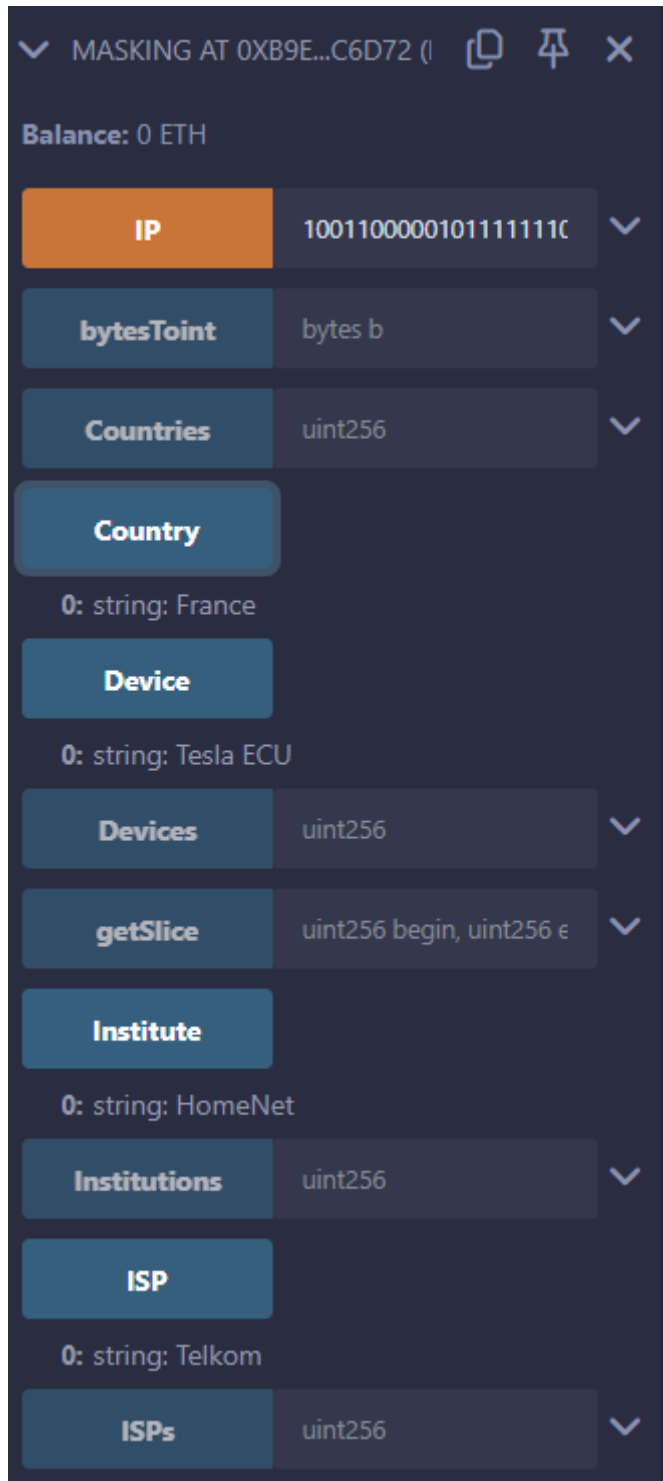
Question 2 - Next, briefly understand how Github uses RSA and SSH to verify the identity of your local computer. You will notice that there is a difference in how public and private keys are used. Briefly explain the difference and under which circumstance each method is appropriate. (1 point)

Private and public key derives from Rivest–Shamir–Adleman cryptographic protocol that rely on a separate set of cryptographic key, one public and one private. The private key is generated using a set of two prime numbers, and the public key is derived from the private key.

SSH on the other hand is cryptographic network protocol that allows for the transmission of data on unsecured network. Unlike RSA keys, SSH uses the private key to prove identity by signing a challenge from the server, while the public key is used by the server to verify that signature.

Therefore, RSA is appropriate whenever there is a need for transmitting data on unsecure means of communication, while SSH is useful for identify authentication on computer networks. If two computers would like to transfer data, they would rather use SCP protocol

Question 3 - Code snapshot



Question 4 - **Blockchain Networking**

The case of students relaying messages on a blockchain illustrates how information can be transmitted but mostly deformed within a network owing to time constraints and rewards mechanisms. Furthermore, this illustrates how malicious or unreliable can degrade data transmissions.

Indeed, in all simulations, students acted as nodes with defined roles, and therefore required to transmit information to others, under different constraints either in time or order of participation. Therefore, in most games, except 1 and 3, each node acted as a decentralized network node, where most nodes could freely choose with which people had to exchange information. Therefore, in most game, all students have typical role of “trustless nodes” (Wang et al., p.2) with low messaging overhead. Furthermore, most final answer given during individual rewards mechanism were correct, which therefore can hint that no person has maliciously altered the information even if he couldn't benefit it himself.

Yet, the main failure resulted in the time limit and number of copies, with detrimental consequences for the retention and reproduction of information. From the deformation of information that was witnessed in my close vicinity during the game, most information was similarly replicated across all people and therefore could hint that most nodes were honest (no extreme deviation from the rest of the node), with reliability hampered only by time constraints. Some unreliability points remained in the different writing style, which could complicate distribution of informations,

However, on round 1, there was a clear sequential information sharing structure, therefore entailing some hierarchy in the nodes access to information, with one node being reliant on the previous one to obtain information. Thus, there was an establishment of hierarchy between students, with some not able to challenge information received from all participating nodes. Most information was corroborated with each other's, with final transmission of data being correct most of the time, except in fourth round where information was incorrectly given owing to lost information, probably due to the time and replication constraints.

Thus, even when rewards were individual, most students acted as honest nodes but were somewhat unreliable owing to time and replication constraints. Although the distribution of node roles was well distributed among people, the architecture and rewards mechanism experimented in sequential games weren't.

First, several different information transmission mechanisms were experimented with throughout the game. The first round followed a line network topology, in which information was passed sequentially from one node to the next. This structure creates a single point of failure at every transmission step: if any individual node fails to reproduce the information accurately, all subsequent nodes will fail. As a result, this configuration represents the highest risk, since errors can easily propagate and cascade through the entire network.

Despite this risk, the line network resembles aspects of blockchain designs like Bitcoin where information is stored in a single chain of blocks (Nakamoto, 2008). In such systems, correctness depends heavily on nodes following the protocol honestly, but Bitcoin achieving reduced risk with continuous verification mechanisms, which allow nodes to independently verify the validity of information before accepting it.

However, in most other rounds, the transmission of information aligned with decentralized network architecture, where each student could easily communicate with any node of his choice. This was particularly salient for round 2 and 3, with similar transmission architecture but different constraints, with round 2 being the slowest-paced, that allowed for reliable information replication.

One interesting network architecture was for round 4, where specific students acted as a bridge node between groups of students and could communicate. This is somewhat similar to “sharded architecture”, where one network of master nodes communicates information between them, with information transmitted within ‘shard’ of nodes between the master nodes and the attached nodes (Yu et al., 2020)

Therefore, most architecture resembled real-life blockchain architectures, except first round that did not abide by information sharing standards.

However, in terms of rewards mechanisms, two main types have emerged. First, during the first four rounds, traditional Proof-of-Work (PoW) (Nakamoto, p.4) mechanism could be identified, with the fastest student reaping the rewards if he answered correctly. Yet, this entailed a perverse mechanism, as there was nothing at stake, and the student was not penalized if he didn’t answer correctly. Even if real-life PoW entails punishment by loss of derivative stakes (electricity or hardware usage), in the case of the class experiment there were no punishments for failure.

Subsequent rounds introduced staking mechanisms (PoS) that entailed the students to stake against the risk of a bad answer, or even collectively stake points against the good answer (Asif & Assan, p.2). Last round introduced physical task, with tied “physical staking” to the total amount of the reward. These rounds were similar to current Ethereum blockchain mechanism.

Thus, in terms of real-life resemblance between the game experiment and current blockchain technologies, rewards mechanisms were the most similar, with PoS and PoW, with students either competing to find the best answer to collect the reward or staking points.

In conclusion, we can therefore witness two points of failure that are particularly damageable for information transmissions and divulgations: Chain of relay and punishment in case of failure

First, these experiments have shown that information chains most susceptible to failure are the ones where information cannot be contested by different sources. Therefore, one clever approach to avoid such bottleneck of information would be to allow for information verification with other nodes. In other words, blockchain protocol should allow for decentralized network, albeit it entails the hypothesis that most nodes participating are honest. This is particularly salient from the class experiment, with most architecture successful following a decentralized architecture

Furthermore, in terms of rewards for participation, the most efficient way to reward participants would be to implement staking protocols, as they would entail sharing risks of dishonesty or error between each node staking, thus facilitating the rise of honest behaviors. This has been seen from the class experiment with students being very attentive to answer, notably when collectively staking, with participants not trusting the nodes not staking participants.

Bibliography:

¹ Wang, W., Hoang, D. T., Hu, P., Xiong, Z., Niyato, D., Wang, P., ... & Kim, D. I. (2019). A survey on consensus mechanisms and mining strategy management in blockchain networks. *Ieee Access*, 7, 22328-22370.

² Fan, C., Ghaemi, S., Khazaei, H., & Musilek, P. (2020). Performance evaluation of blockchain systems: A systematic survey. *Ieee Access*, 8, 126927-126950.

³ Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system.

⁴ Yu, G., Wang, X., Yu, K., Ni, W., Zhang, J. A., & Liu, R. P. (2020). Survey: Sharding in blockchains. *IEEE Access*, 8, 14155-14181.

⁵ Asif, R., & Hassan, S. R. (2023). Shaping the future of Ethereum: Exploring energy consumption in Proof-of-Work and Proof-of-Stake consensus. *Frontiers in Blockchain*, 6, 1151724.